Chapter 4
Magnetic Fields and Magnetic Forces

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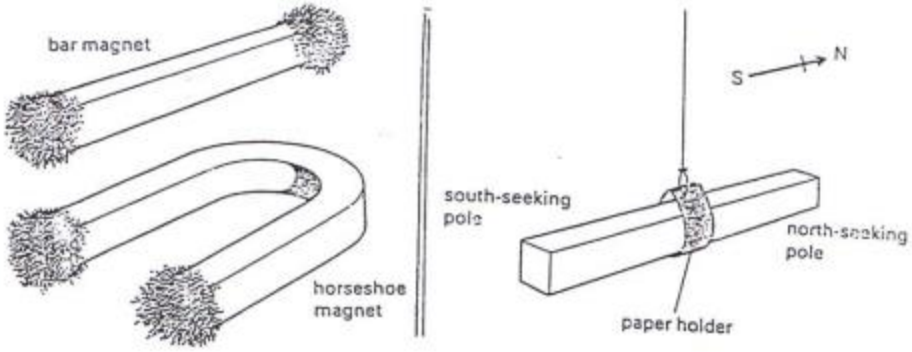
MAGNETIC FIELDS AND MAGNETIC FORCES

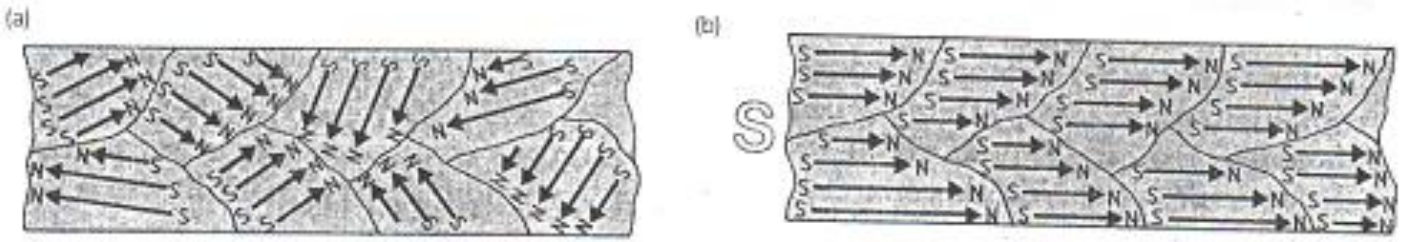
Properties of Magnet

- 1- A magnet attracts iron pieces or iron fillings. The filings cling near the ends, at the (poles) of the magnet.
- 2- The magnetization is zero at the middle of the magnet and maximum at the poles.
- 3- When a magnet is freely suspended so that it can swing in a horizontal plane, it comes to rest in N-S direction. N-pole points towards the north, and S-pole points towards the south.



- 4- First law of magnetism: " like poles repel, unlike poles attract".
- 5- When a magnet is broken into pieces, each piece is found to be a magnet with two poles, i.e. no monopole magnet.

<u>What causes magnetism</u>? It is the spinning of the electrons in the atoms of materials, and the behavior of these electrons, which give a material its magnetic properties.



Domains in (a) an unmagnetised object. (b) a magnetised object

<u>Magnetic field</u>: "It is the region around a magnet in which a magnetic force is exerted". We can describe magnetic interactions in two steps:

1- A moving charge or a current creates a magnetic field in the surrounding space (in addition to its electric field)

2- The magnetic field exerts a force $\vec{F}$ on any other moving charge or current that is present in the field.

Magnetic field is a vector quantity and denoted by $\vec{B}$. At any position the direction of $\vec{B}$ is defined as the direction in which the north pole of a compass needle tends to point.

The Magnetic Force on a Moving Charge: The magnetic force, $\mathbf{F}_B$, on a test charge, q_o , moving with velocity v in a magnetic field, is

$$\vec{F} = q\vec{v} \times \vec{B}$$

The magnitude of the magnetic force is $|F| = q\,v\,B\sin\phi$, where ϕ is the angle measured from the direction of v to the direction of **B** . The force is perpendicular both to the velocity and to the magnetic field. Since the force is proportional to the velocity, charges at rest do not experience magnetic forces.

$$[B] = T\ (\text{Tesla}) = \frac{N}{A.m}\ .\ \text{Also, } 1T = 10^4\ \text{Gauss.}$$

Magnetic Field Lines and Magnetic Flux

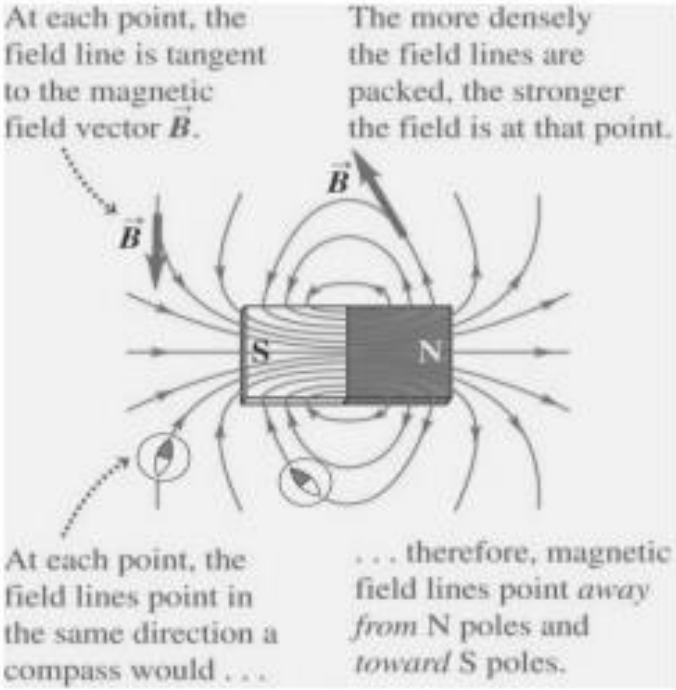
The magnetic flux lines or lines of force represent the magnitude and direction of the magnetic field, i.e. it is vector quantities.

``Note that:

1- `The density of lines is proportional to the strength of the field (i.e. the lines are close together where the field is strong and vice versa).

2- The lines go from North Pole to the South Pole.

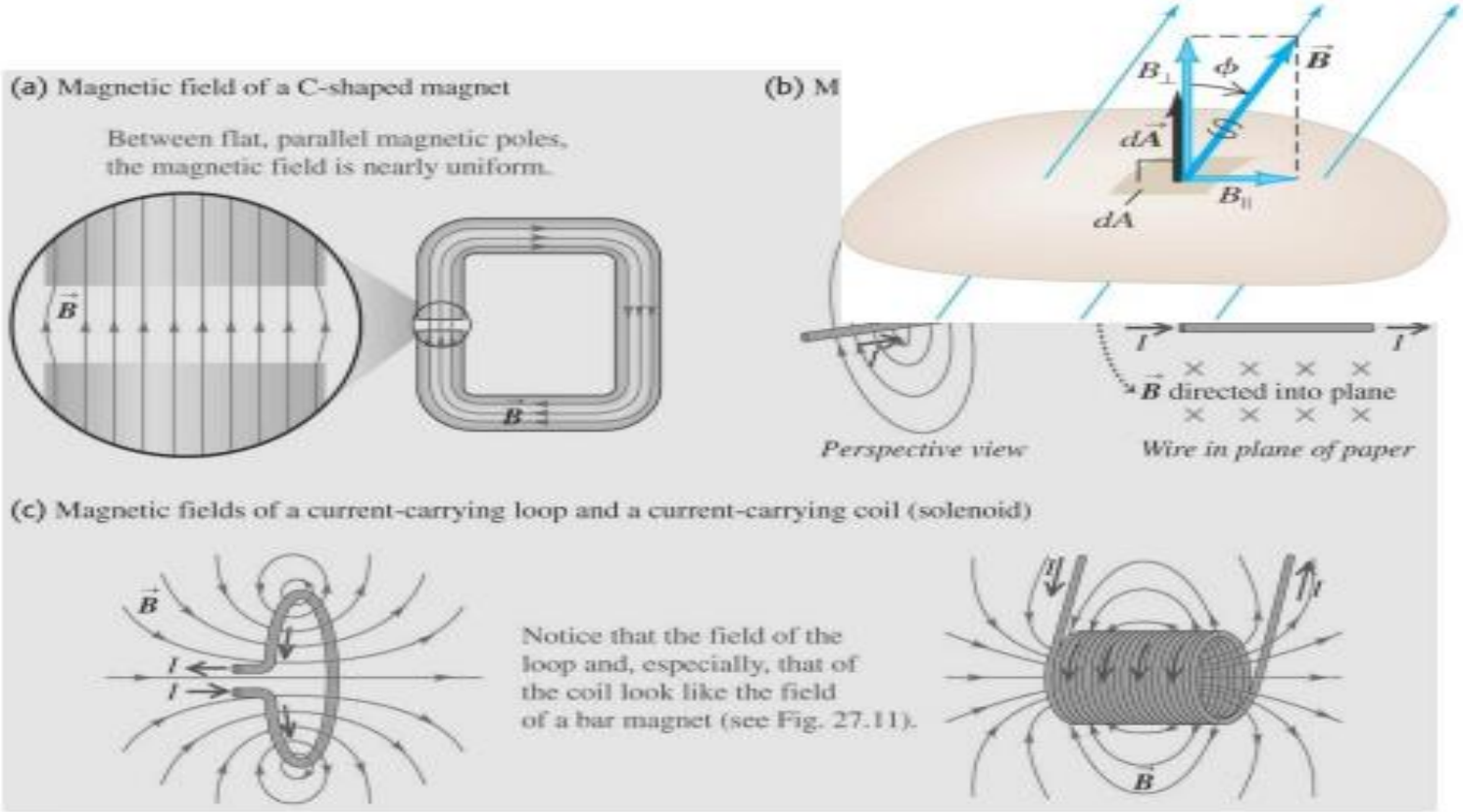
3- The lines never cross each other.



Magnetic Flux and Gauss's Law for Magnetism

We can divide any surface into elements of area dA. For each element we determine $B_\perp$ the component of $\vec{B}$ normal to the surface at the position of that element. We define the magnetic flux $d\phi_B$ through this area as

$$d\phi_B = B_\perp dA = B\cos\varphi dA = \vec{B}.d\vec{A}$$



Where φ is the angle between the direction of $\vec{B}$ and the line perpendicular to the surface. The total magnetic flux through the surface is the sum of the contributions from the individual area elements:

$$\phi_B = \int B_\perp dA = \int B\cos\varphi dA = \int \vec{B}.d\vec{A}$$

Magnetic flux is a scalar quantity. In the special case in which $\vec{B}$ is uniform over a plane surface with total area A, $B_\perp$ and φ are the same at all points on the surface, and

$$\phi_B = B_\perp A = BA\cos\varphi$$

The SI unit of magnetic flux is called **weber (Wb)**.

$$1\ Wb = 1\ T.m^2 \quad ,1\ T = 1N/A$$

Gauss's law for magnetism

The total magnetic flux through a closed surface is always zero. Symbolically

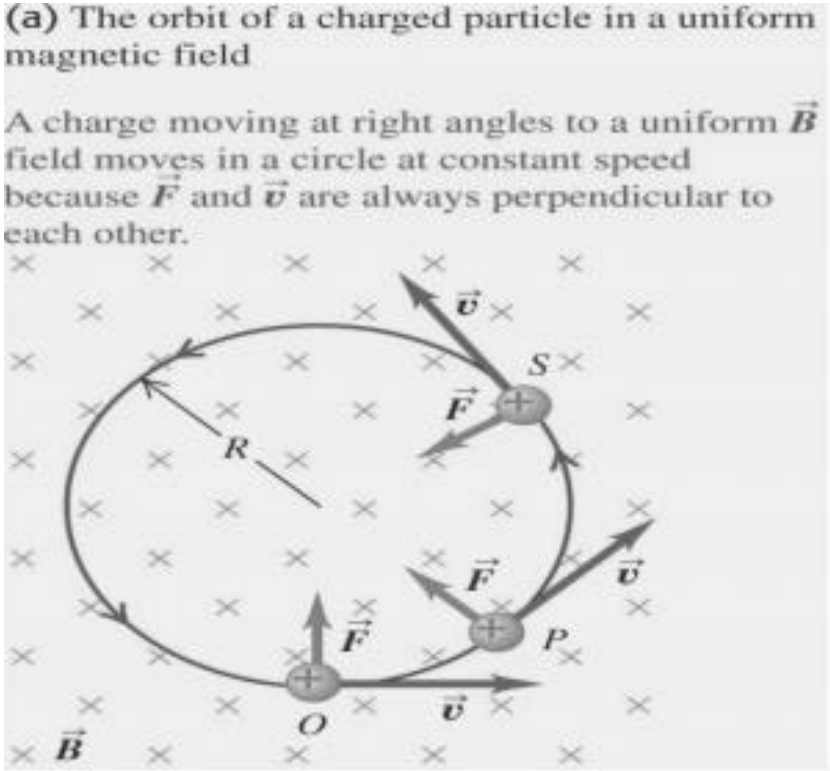
$$\oint \vec{B}.d\vec{A} = 0$$

If the element of area dA is at right angles to the field lines, then $B_\perp = B$; calling the area $dA_\perp$, we have

$$B = \frac{d\phi_B}{dA_\perp}$$

That is, the magnitude of magnetic field is equal to flux per unit area across an area at right angles to the magnetic field. For this reason, magnetic field B is sometimes called **magnetic flux density.**

<u>A Beam of Charged Particles in a Magnetic Field</u>: Consider a beam of positively charged particles with velocity v .If this beam enters a magnetic field at right angle to its direction of motion, it will experience a force perpendicular to both velocity and magnetic field, i.e. it will be deflected.



Circular motion: The beam will move in circular path if the velocity is small and the magnetic field is strong. In that case: Magnetic force = Centripetal force

$$qvB = m\frac{v^2}{R} \quad \Rightarrow \quad B = \frac{mv}{qR}$$

where R is the radius of the circle.

Also, the angular frequency (ω) and the periodic time (T) of a rotating charged particle are:

$$\omega = \frac{v}{R} = \frac{\frac{qBR}{m}}{R} = \frac{qB}{m}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi m}{qB}$$

Application of motion of charged particles

Velocity selector: In a region of crossed or perpendicular magnetic field B and electric field E perpendicular to v, the forces cancel when,

$$\boxed{qvB = qE \Rightarrow v = \frac{E}{B}}$$

- Only particles with speeds equal to E/B can pass through without being deflected by the fields.

The Magnetic Force on a Current Carrying Wire:

- The magnetic force $\mathbf{F}_B$ on a segment of wire, **L** , carrying current, I, in a magnetic field, **B** , is given by:

$$\boxed{\boldsymbol{F}_B = q_o \mathrm{v} \times \boldsymbol{B} = q_o \frac{\boldsymbol{L}}{\mathrm{t}} \times \boldsymbol{B} = \frac{q_o}{t} \boldsymbol{L} \times \boldsymbol{B} = I\left(\boldsymbol{L} \times \boldsymbol{B}\right)}$$

- The total force on the wire is the vector sum of the forces on the segments.

Examples

Example1: An electron is projected into a uniform magnetic field given by

B =(1.4 ˆi + 2.1 ˆj) T. Find the vector expression for the force on the electron when its velocity is **v** =3.7 x10^5 ˆj m/s .

Answer:

$$F_B = q_e \mathrm{v} \times \boldsymbol{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ v_x & v_y & v_z \\ B_x & B_y & B_z \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 3.7\times10^5 & 0 \\ 1.4 & 2.1 & 0 \end{vmatrix} = q_e(1.4)(3.7\times10^5)(-\hat{k}) = \underline{(8.3\times10^{-14}\hat{k})\ \mathrm{N}}$$

Example 2: A proton is moving with a velocity of **v** = 6.0 x 10^6 iˆ m/s at a point where the magnetic field is given by **B** =(3.0 ˆi – 1.5 ˆj + 2.0 ˆk) T . What is the magnitude of the acceleration of the proton at this point?

Answer:

$$\because F_B = m\boldsymbol{a} = q_e \mathrm{v} \times \boldsymbol{B} \therefore \boldsymbol{a} = \frac{q_e}{m}\mathrm{v} \times \boldsymbol{B} = \frac{1.6\times10^{-19}}{1.67\times10^{-27}} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 6.0\times10^6 & 0 & 0 \\ 3.0 & -1.5 & 2.0 \end{vmatrix} :$$

$$= -1.2\times10^7\hat{j} - 9.0\times$$

$$\therefore |\boldsymbol{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2} = \underline{1.44\times10^{15}\ \mathrm{m/s^2}}$$

Example 3: An electric field and a magnetic field normal to each other. The electric field is 4.0 kV/m and the magnetic field strength is 2.0 mT. They are act on a moving electron to produce no force, a- Calculate the electron speed. b- Calculate the kinetic energy of the electron.

Answer:

a-

$$v = \frac{|E|}{|B|} = \frac{4.0\times10^{3}}{2.0\times10^{-3}} = \underline{2.0\times10^{6}\ \text{m/s}}$$

b-

$$k = \frac{1}{2}mv^{2} = \frac{1}{2}(9.11\times10^{-31})(2.0\times10^{6})^{2} = \underline{1.82\times10^{-18}\ \text{J}} = \underline{11.4\ \text{eV}}$$

H.W. A proton with velocity **v** = 2.0 x 10^6 ˆi (m/s) moves horizontally into a region of space in which there is an electric field **E** = -5.0 x10^3 ˆj N/C and a magnetic field B. Find the smallest magnetic field such that the proton will continue to move horizontally, i.e. un-deflected. [Ans: **B** = -2.5 x10^{-3} ˆk T]

Example 4: A proton with a velocity of 6.0×10^6 m/s Travels at right angles to magnetic field of 0.5 Tesla. What is the frequency of the proton's orbit?

Answer:

$$\because qvB = m\frac{v^{2}}{r} \therefore \frac{v}{r} = \frac{qB}{m} \Rightarrow f = \frac{\omega}{2\pi} = \frac{v}{2\pi r} = \frac{qB}{2\pi m} = \frac{1.6\times10^{-19}\times0.5}{2\pi\times1.67\times10^{-27}} = \underline{7.62\times10^{6}\ \text{Hz}}$$

Magnetic Dipoles: A current loop, of area A, behaves like a magnet and has a magnetic dipole moment

$$M = IA\hat{n}$$

Where $\hat{n}$ is a unit vector normal to the plane of the loop. Its direction is found by placing the **fingers** of the right hand along the **current**; the **right thumb** then indicates the direction of $\hat{n}$ and $\boldsymbol{M}$. For N –current Loops.

$$M = NIA\hat{n}$$

In a **uniform magnetic field**, a magnetic dipole behaves much like an electric dipole in uniform electric field, and the net force on the dipole will be zero. The torque, , on the dipole and the potential energy, U, are given by:

$$\tau = \boldsymbol{M} \times \boldsymbol{B}, \qquad U = MB\cos\theta$$

The magnetic torque on a flat current-carrying loop of wire by a uniform magnetic field is maximum when the plane of the loop is parallel to B.

Example:

The plane of area 4.0 cm^2 rectangular loop of wire is parallel to a 2.0 T magnetic field. The loop carries a current of 6.0 A. Calculate the magnitude of the torque acts on the loop.

Answer:

$$\tau = \boldsymbol{M} \times \boldsymbol{B} = NIA \times \boldsymbol{B} = NIAB\sin\theta$$

$$= (1)(6)(4\times10^{-4})(2)\sin 90^{\circ} = 4.8\times10^{-3}\ \text{N.m}$$